

The Fixed-ONB Banded Operator Algebra

FA/Banach run packet for arXiv:1408.1164

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Status

This packet gives a likely valid full answer to the second open question in Remark 1.92 of Jorgensen–Tian [1]. It does not address the first, intrinsic-geometric question about which operators admit some banding orthonormal basis.

Source Question

The source asks, after fixing an orthonormal basis $(e_i)_{i \in \mathbb{N}}$ and

$$\mathcal{D} = \text{span}\{e_i : i \in \mathbb{N}\},$$

what the $*$ -algebra is of unbounded operators with dense domain $\mathcal{D}$ whose matrices are banded with respect to (e_i) .

The question appears in Remark 1.92, PDF pages 73–74. The source crops are included below.

$$\begin{aligned} &= \sum_k \langle A^* u_i, u_k \rangle \langle u_k, B u_j \rangle \\ &= \langle A^* u_i, B u_j \rangle \quad [\text{by Parseval}] \\ &= \langle u_i, A B u_j \rangle \\ &= (M_{AB})_{ij} \end{aligned}$$

where we used that $I = \sum |u_i\rangle\langle u_i|$. □

Exercise 1.91 (Matrix product of $\infty \times \infty$ banded matrices). Consider two linear operators A and B both defined on a dense subspace $\mathcal{D}$ in a fixed Hilbert space $\mathcal{H}$. Suppose $\mathcal{D}$ contains an ONB $\{e_i\}_{i \in \mathbb{N}}$, and that the corresponding matrices M_A and M_B with respect to $\{e_i\}$ are both banded. Then show that the matrix-product

$$M_{AB} = M_A M_B \tag{1.60}$$

is well defined, and is again banded. See Figure 1.14.

Hint: Use Lemma 1.90, and the equation

$$(M_{AB})_{i,j} = \langle e_i, A B e_j \rangle = \langle A^* e_i, B e_j \rangle,$$

and note that $\mathcal{D} \subset \text{dom}(A^*)$.

Remark 1.92. Here are two *open questions* regarding *banded* operators/matrices.

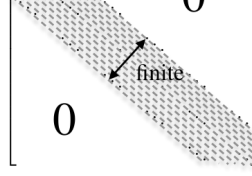


Figure 1.14: $\infty \times \infty$ banded matrix. Supported on a band around the diagonal.

1. Fix a separable Hilbert space $\mathcal{H}$, is there an intrinsic geometric characterization of the linear operators A (with dense domain) in $\mathcal{H}$ which admit an ONB $\{e_i\}_{i \in \mathbb{N}}$ such that the matrix

$$(MA)_{i,j} := \langle e_i, Ae_j \rangle \quad (1.61)$$

is banded?

2. Given an ONB $\{e_i\}_{i \in \mathbb{N}}$, what is the $*$ -algebra $\mathfrak{A}$ of unbounded operators with dense domain

$$\mathcal{D} = \text{span} \{e_i\} \quad (1.62)$$

such that every $A \in \mathfrak{A}$ is banded with respect to $\{e_i\}_{i \in \mathbb{N}}$?

Exercise 1.93 (A transform). Let $\mathcal{H}$ be a Hilbert space, and A a set which indexes a fixed ONB $\{v_\alpha\}_{\alpha \in A}$. Now define $T : \mathcal{H} \rightarrow l^2(A)$, by

$$(Th)(\alpha) := \langle v_\alpha, h \rangle, \quad \forall \alpha \in A, h \in \mathcal{H}.$$

Show that T is unitary, and onto $l^2(A)$, i.e.,

$$\begin{aligned} TT^* &= I_{l^2(A)}, \text{ and} \\ T^*T &= I_{\mathcal{H}}. \end{aligned}$$

T is called the analysis transformation, and T^* the synthesis transformation.

Three Norm-Completions

Let $\mathcal{H}$ be a fixed Hilbert space, infinite-dimensional in the discussion below.

Proof Intuition

Once the orthonormal basis and common core are fixed, there is no hidden operator-theoretic obstruction. A banded matrix has finite propagation along the index set. Finite propagation is preserved by conjugate transpose, and propagation constants add under multiplication. Therefore the maximal common-core $*$ -algebra is exactly the algebra of all finite-propagation matrices on $\mathcal{D}$; any smaller answer is merely a $*$ -subalgebra of it.

The Result

Let $\mathcal{H}$ be a separable Hilbert space with fixed orthonormal basis $(e_i)_{i \in \mathbb{N}}$, and set $\mathcal{D} = \text{span}\{e_i : i \in \mathbb{N}\}$. For a linear operator $A : \mathcal{D} \rightarrow \mathcal{D}$, write

$$a_{ij} = \langle e_i, Ae_j \rangle.$$

Say that A has bandwidth at most r if $a_{ij} = 0$ whenever $|i - j| > r$.

Theorem 1. *The maximal common-domain $*$ -algebra of operators on $\mathcal{D}$ whose matrices are banded with respect to (e_i) is*

$$\mathcal{A}_{\text{band}}(\mathcal{D}) = \bigcup_{r \geq 0} \{A : \mathcal{D} \rightarrow \mathcal{D} : \langle e_i, Ae_j \rangle = 0 \text{ whenever } |i - j| > r\}.$$

The involution is the Hilbert-space adjoint restricted to $\mathcal{D}$, equivalently conjugate transpose of the matrix. Moreover, if $\mathfrak{A}$ is any common-domain $*$ -algebra on $\mathcal{D}$ all of whose elements are banded with respect to (e_i) , then $\mathfrak{A} \subseteq \mathcal{A}_{\text{band}}(\mathcal{D})$.

Proof. First let $A \in \mathcal{A}_{\text{band}}(\mathcal{D})$ have bandwidth at most r . For each basis vector e_j , the vector Ae_j is a finite linear combination of e_i with $|i - j| \leq r$. Hence A maps $\mathcal{D}$ into $\mathcal{D}$ and is densely defined.

For fixed i , the functional $x \mapsto \langle e_i, Ax \rangle$ on $\mathcal{D}$ is represented by the finite vector

$$\sum_{|j-i| \leq r} \overline{a_{ij}} e_j.$$

Thus $e_i \in \text{dom}(A^*)$ for every i , so $\mathcal{D} \subseteq \text{dom}(A^*)$, and the restricted adjoint $A^\dagger := A^*|_{\mathcal{D}}$ maps $\mathcal{D}$ into itself. Its matrix is $a_{ij}^\dagger = \overline{a_{ji}}$, so it also has bandwidth at most r .

If A and B have bandwidths at most r and s , respectively, then $A + B$ has bandwidth at most $\max\{r, s\}$. For the product, the matrix entry of AB is

$$(AB)_{ij} = \sum_{k \in \mathbb{N}} a_{ik} b_{kj}.$$

The sum is finite, since both matrices are banded. If $|i - j| > r + s$, then no index k can satisfy both $|i - k| \leq r$ and $|k - j| \leq s$, so every summand vanishes. Therefore AB has bandwidth at most $r + s$. Scalars and the identity are immediate, and $\mathcal{A}_{\text{band}}(\mathcal{D})$ is a unital common-core $*$ -algebra.

Finally, suppose $\mathfrak{A}$ is any $*$ -algebra of operators with common domain $\mathcal{D}$ such that every element is banded with respect to (e_i) . By the definition of bandedness, each $A \in \mathfrak{A}$ has some finite bandwidth, and therefore $A \in \mathcal{A}_{\text{band}}(\mathcal{D})$. Hence $\mathfrak{A} \subseteq \mathcal{A}_{\text{band}}(\mathcal{D})$. This proves maximality and also explains the mild ambiguity in the source wording: without the maximality convention, every $*$ -subalgebra of $\mathcal{A}_{\text{band}}(\mathcal{D})$ is also an answer. $\square$

Verification Notes

The proof is purely algebraic. The included script `code/verify_bandwidth_arithmetic.py` checks the bandwidth arithmetic on finite truncations: adjoints preserve bandwidth and products have bandwidth bounded by the sum of bandwidths. This sanity check is not used as evidence for the proof, but it is a useful guard against an indexing mistake.

Local novelty checks were made against the run indexes for arXiv:1408.1164 and for phrases including `banded operator`, `finite propagation`, `finite band`, `band-dominated`, and `common domain`. No exact local duplicate was found. A web search on 2026-07-03 found adjacent band-dominated-operator literature, but no exact local packet for this fixed-ONB common-core algebra question. Since the answer is elementary, it should be regarded as likely folklore even though it resolves the literal question as stated.

Scope

This packet answers only Remark 1.92, Question 2, after fixing the ONB and the common dense core $\mathcal{D}$. It does not give an intrinsic geometric characterization of all densely defined operators which admit some banding ONB, which is the first source question.

References

- [1] Palle Jorgensen and Feng Tian, *Noncommutative analysis, Multivariable spectral theory for operators in Hilbert space, Probability, and Unitary Representations*, arXiv:1408.1164.